Computer Science and Engineering Department, IIIT Kota

End Term Examinations, Nov 2024

M.Tech (Semester-I)

Subject: **Statistics for Data Science** — Subject Code: **CST501**

Duration: **2 Hours** — Full Marks: **40**

Answer all questions. All parts of a question must be answered in sequence. Figures at the right margin indicate marks.

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1. (a) What are data quality problems? Discuss different approaches to handle these problems. [3]

   (b) The joint probability distribution of X and Y is given by

   $$f(x,y) = \frac{x+y}{21};\ x = 1,2,3;\ y = 1,2$$

   Find (i) the marginal distribution of X and Y, (ii) the expectation of X and XY. [3]

   (c) Write the importance of feature selection and also discuss one filter-based technique widely used for selection of features. [4]

2. (a) For the following two-dimensional dataset:

   Class 1 points: (2,1), (3,5), (4,3)

   Class 2 points: (5,6), (6,7), (7,8)

   determine its lower dimensional representation (1D) using principal component analysis. Analyze the step-wise results. [5]

   (b) How does the complexity of a model relate to bias and variance? List a few techniques to lower variance in a model without increasing bias. [3]

   (c) You are given a dataset with three numerical features and one categorical feature, and your task is to perform a classification task. It is observed from the dataset that one numerical feature and one categorical feature have a few missing values corresponding to a few observations. How would you deal with these missing values before you fed your data to a classifier? [2]

3. (a) Why is data normalization an important task in data science?. [2]

   (b) Consider the following joint distribution of random variables X and Y:

   $$f(x,y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 \le x \le 2,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

   Check whether X and Y are mutually uncorrelated and/or independent. [3]

   (c) Consider a given two-class classification task on a two-dimensional space, where data in both the classes are distributed according to Gaussian distribution with $\mu_1 = [0,0]$, $\mu_2 = [3,3]$, and

   $$\Sigma_1 = \Sigma_2 = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

   Assuming $P(c_1) = P(c_2) = \frac{1}{2}$, classify the data point $x = [1.0, 2.2]$ into class c_1 or c_2 according to minimum distance classifier. [3]

   (d) What are Type I and Type II errors? Explain with suitable examples. [2]

4. (a) What is cross-entropy? Calculate the entropy in bits for the random variable indicating pixel values in an image whose possible intensity values are 0 and 1 with uniform probability. [4]

   (b) What are t-tests and p values? Why do we need these tools in data science? [3]

   (c) Prove that $H(X,Y) = H(X) + H(Y|X)$, where $H(X,Y)$ denotes the joint entropy of two random variables X and Y. [3]

$\int x \cdot f(x)\, dx$

********** All the Best **********

IIIT Kota

Department of Computer Science and Engineering

CST505 Machine Learning: Tools and Techniques End Term 2024 MM: 40

Time: 2 hrs

Note: Write to the point with mathematical justifications.

1. Consider a perceptron to represent the Boolean function AND with the initial weights $w_1 = 0.3$ and $w_2 = -0.2$, learning rate $\eta = 0.2$ and bias/threshold $(b = 0.4)$. The activation function used here is the step function $f(x)$ which gives the output as 0 or 1. If the value of $f(x) = (x_1w_1 + x_2w_2 - b \geq 0)$, it outputs 1 else it outputs 0. Design a perceptron that performs the Boolean function AND and update the weights for two epochs only. (10)

2. Consider the following table of observations to assess a students's performance during his course of study to predict whether the student will get a scholarship offer or not. Compute the Gini index of the Interactive attribute only. (5)

No	Interactive	Programming Skills	Presentation skills	Scholarship offer
1.	yes	very good	good	no
2.	no	good	moderate	yes
3.	no	average	Poor	no
4.	yes	average	good	no
5.	yes	good	moderate	yes
6.	yes	good	moderate	yes
7.	yes	very good	poor	no
8.	no	very good	good	no
9.	yes	good	good	yes
10.	yes	average	good	yes

3. Perform k-means clustering on the data below. Set $k = 2$ and the initial value of objects 2 and 5 with the coordinate values $(6, 4)$ and $(4, 12)$ as initial seeds. Show the values of centroids and the distances from samples to centroids for the first two iterations only. Use Euclidean distance metric as the distance measure.(5)

objects	x-coord	y-coord
1.	12	10
2.	6	4
3.	6	8
4.	15	14
5.	4	12

4. Suppose we have a classifier to clasify medical data, where the class label is cancer. The confusion matrix of the model applied to a set of 200 observations is given in table below. Calculate accuracy, precision, recall, sensitivity and specificity. Is the classifier good enough to accuractely recognize both positive and negative samples? (5)

	Predicted +ve	Predicted -ve
Actual +ve	0	7 → 11
Actual -ve	4	185

TP FN Recall TP/TP+FN

FP TN

TP ~~FP~~ / TP+FP

5. Given probability of having the disease is $P(d) = 0.0002$. The likelihood of testing positive given the patient had disease is $P(t|d) = 0.98$. Use Bayes' theorem to calculte i) the probability that the patient actually has the disease based on the evidence of the test result $P(d|t)$. ii) Also calculate the overall probability of the test returning a positive value $p(t)$. (5)

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTA

M.Tech. (CSE) Semester –I

End Term Examination, Odd Semester 2024-25

Nature Inspired Algorithms (CST509)

Marks: 40 | **Time: 120 minutes** | **Date: 04.12.2024**

Note: Each question carries **(10 marks).**

Calculators not allowed.

Q1. Explain the Initialization Phase, Employed Bee Phase, Onlooker Bee Phase and Scout Bee Phase of Artificial Bee Colony Meta Heuristic Algorithm in detail with Pseudo Codes.

Q2. Explain and discuss different kinds of mutation strategies giving example against each.

Q3. With the help of Genetic Algorithm maximize the function $f(x) = x^2$, where value of x ranges from 0-31. Take all necessary assumptions. Take each step of the algorithm into account.

Q4. Write short notes on

(a) Fitness Scaling

(b) Real life applications where NIA can be used

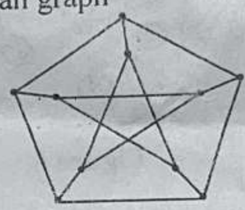
Linear Power Sigma

Department of Computer Science and Engineering, IIIT Kota
End Term Examination 2024, M.Tech (CSE) 1st Semester

Subject: Advanced Algorithms CST507
Date and time:3 Dec, 2024; (10:30am - 12:30pm)
Note: Attempt any FIVE questions. Question No. 1 is compulsory. All questions carry equal marks.
Max Marks:50 **(** ***Distribution:*** **Q1:1x10, Q2:10, Q3:5+5, Q4:10, Q5:10 Q610,Q7:10)**

Q1 Answer whether ***TRUE*** of ***FALSE*** with justifications

(a) If m (# of slots) is proportional to n (# of elements in the table), Searching takes constant time on average
(b) The best case running time for the quick-sort when all the elements are same is $O(n^2)$
(c) If the probability that an algorithm errors is ZERO for at least one of the possible output (Yes or No) is referred to as Monte-carlo algorithm double sided error algorithm
(d) An arbitrary sequence {x1, x2, ..., xn} of n numbers contains repeated occurrences of some number can be determined in $\Theta(n\log n)$ time
(e) The memorized algorithm can solve the matrix-chain multiplication problem in $O(n^2\log n)$
(f) A graph G is Eulerian if and only if it is connected and every vertex of G has odd degree
(g) Minimum weight maximum cardinality matching can be obtained in polynomial time.
(h) The minimum weighted vertex cover of a tree can be computed in linear time
(i) Las Vegas becomes a Monte Carlo on the non zero probability error
(j) The following graph is a Hamiltonian graph



Q2. The Set Cover (SC) generalizes Vertex Cover (VC): In VC, elements (clients) are edges, and sets (servers) correspond to vertices. The set corresponding to a vertex v is the set of edges incident to v. Describe the Approx WSCP by Pricing Method and minimize the weight of set cover C for the set $X = \{x_1, x_2, \dots, x_n\}$ of 8 elements for the family $F = \{S_1, S_2, , \dots, S_m\}$ of 8 subsets of X, and w(F): weights $w(S)>0$, for each $S \in F$ as shown in Figure 2

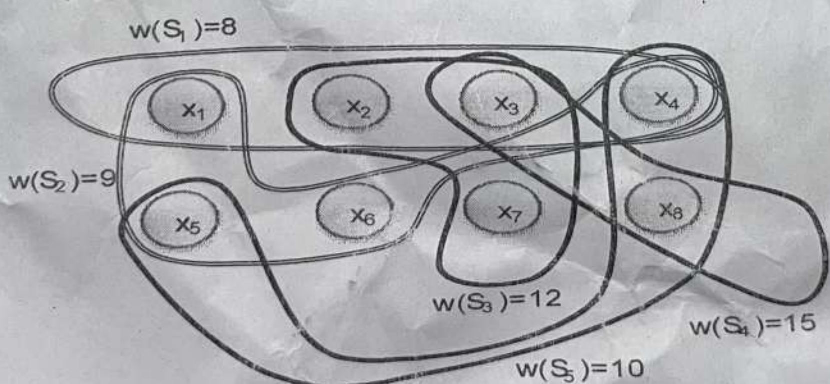


Figure 2

Q3. What are the different methods of hash functions ? Explain with example the hash key generation with the following values using universal hashing function $H_{a,b(k)}$ where, a=3,b=4, K=8, Prime P= 17 and m=6

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M.Tech/PhD (CSE), Semester – I
End Term Examination, Odd Semester 2024-25

Artificial Intelligence (CST503)

Marks: 40 **Time: 120 min** **Date: 05-12-2024**

Note: Attempt all the questions. Take suitable assumptions, if required.

Question 1: Take an example grid i.e. two-dimensional array of integers. Apply Breath-First-Search (BFS) on the grid. Explain all the steps clearly. Also analyze its time complexity. [5 marks]

Question 2: Consider the following 8-puzzle game problem in AI. Apply A* search algorithm to solve it. Show at least 2 iterations. Take any suitable heuristic distance. [5 marks]

Initial State

1	2	3
8		4
7	6	5

Goal State

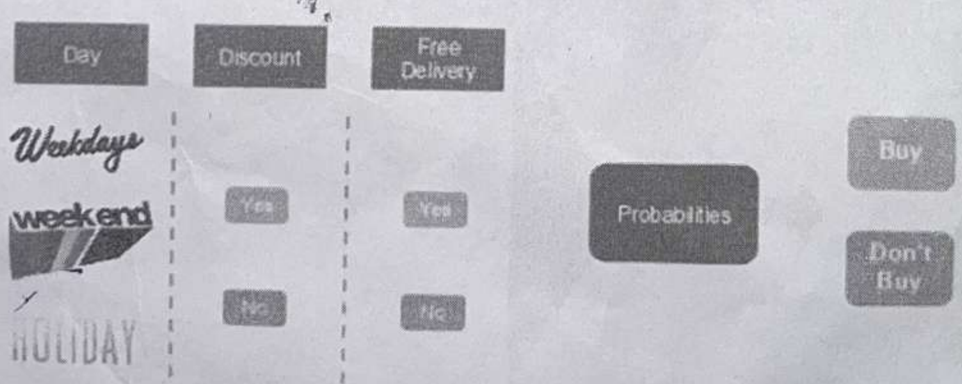
2	8	1
	4	3
7	6	5

Question 3: Take any real-time situation. Solve it using hill-climbing algorithm and simulated annealing. Write your observations. [5 marks]

Question 4: Explain in details any one of the following architecture for object detection:
RCNN or Fast RCNN or Faster RCNN [10 marks]

Question 5: What are the differences between one-stage and two-stage object detection models? Discuss in details. [5 marks]

Question 6: Let's take a shopping mall case study. Suppose we have a small dataset containing three attributes 'Day', 'Discount' and 'Free Delivery' with some permissible values as below:



Create your own dataset of 20 rows having values in any combinations. Using Bayes Theorem, calculate the probability of not making a purchase by a customer, no matter what day of the week it is. [10 marks]